

Tableau

NIT Trichy

Tableau

- Tableau is a **visual analytics platform** transforming the way we use data to solve problems—empowering people and organizations to make the most of their data.
- Tableau was founded in 2003 as a result CS project at Stanford to **improve the flow of analysis and make data more accessible to people through visualisation.**
- Co-founders: **Chris Stolte, Pat Hanrahan and Christian Chabot.** They developed and **patented** Tableau's foundation technology, **VizQL**-which visually expresses data by translating drag-and-drop actions into data queries through an intuitive interface.
- Tableau was acquired by **Salesforce** in 2019.

Tableau

- Connection to various data sources and share the reports/dashboards with all the members in your team or organization.
- Tableau offers drag and drop features. Minimal coding unlike Dash Plotly python library.

Connect to Data

- Tableau provides connections to various data sources.
- Data extract and Real-time data connections.
- Full control over how data are connected. Filter data from source, Load data when needed, real-time data refresh and, optimized data loading etc.

Connect to Data

Example: Sample Superstore Excel File.

In Data source pane, all the worksheets in the file are listed. Each worksheet is a considered as Table.

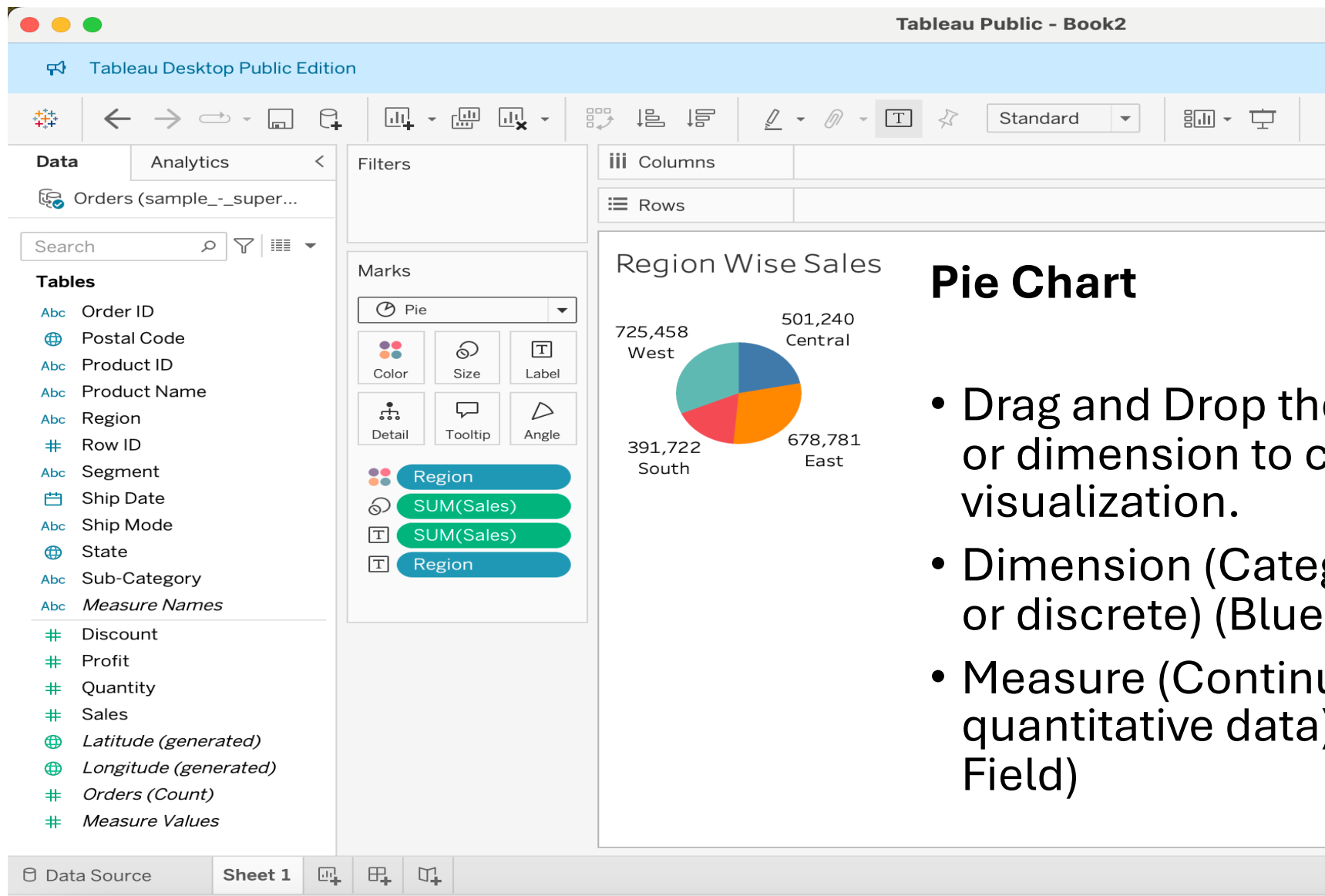
Drag and Drop the needed tables to right pane and establish the relationship among the tables (Join Tables).

Connect to Data

Example: Sample Superstore Excel File.

Orders Worksheet:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
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Pie Chart

- Drag and Drop the measure or dimension to create visualization.
- Dimension (Categorical Data or discrete) (Blue Field)
- Measure (Continuous or quantitative data) (Green Field)

Create Custom Field (Calculation Field)

Order Profitable?

×

```
/*  
Calculate profit at order level  
If greater than zero, it is profitable.  
*/  
  
{ FIXED [Order ID]: SUM([Profit]) } > 0
```

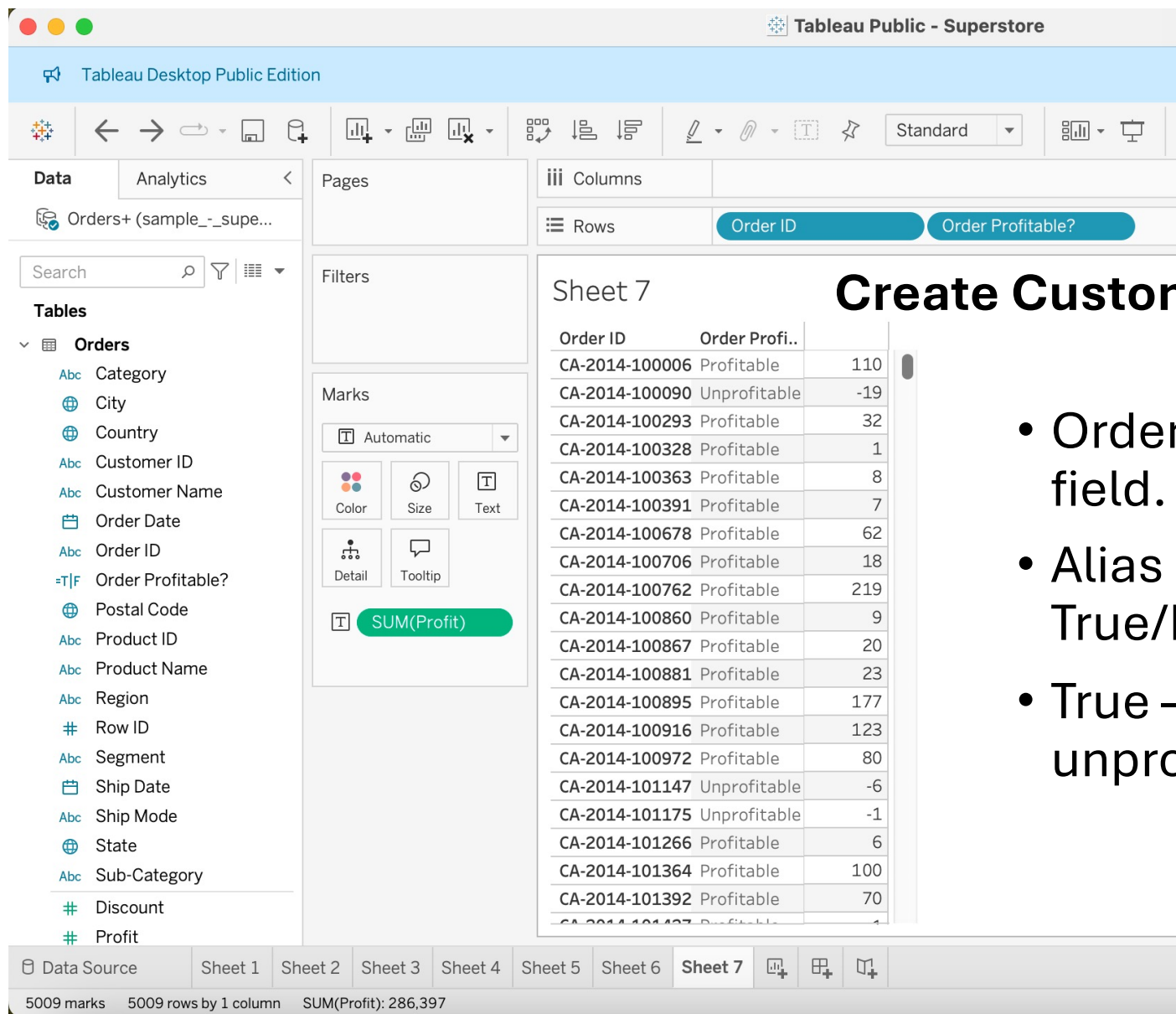
The calculation is valid.

2 Dependencies ▼

Apply

OK

- LOD (Level of Detail)
Expressions: Calculations at different level of granularity
- FIXED LOD: To create a calculated field that aggregate data using a specific set of dimensions.
- If $\text{Sum}(\text{profit}) > 0$, TRUE, else FALSE.



Create Custom Field (Calculation Field)

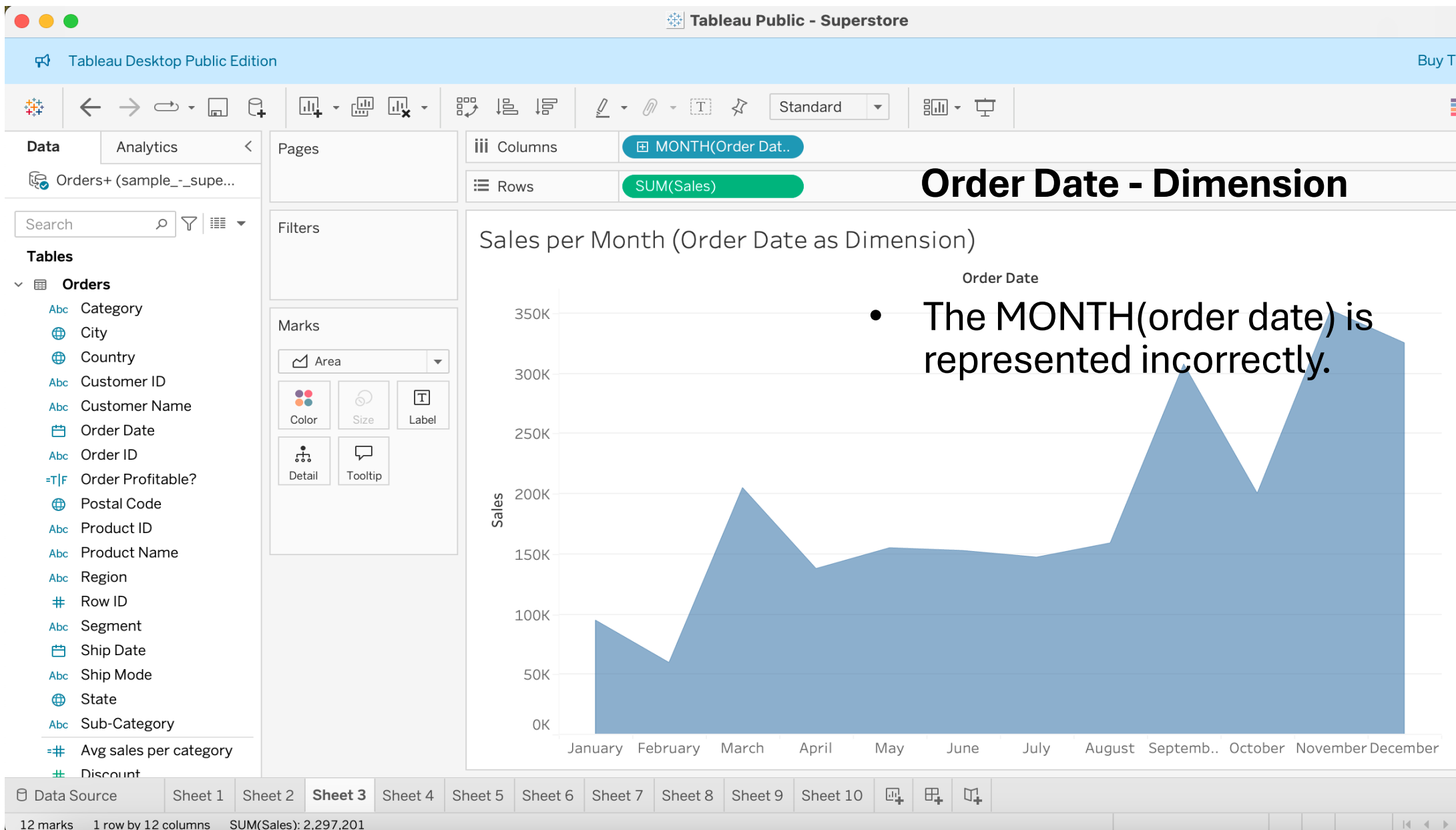
- Order Profitable is a new field.
- Alias is created for the values True/False.
- True – profitable, False – unprofitable.

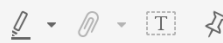
LOD: FIXED, INCLUDE, EXCLUDE

FIXED: Used when you need to isolate specific dimensions for calculations, regardless of the view's dimensions or filters.

INCLUDE: Use when you want to factor in additional dimensions dynamically along with the view's dimensions.

EXCLUDE: Use when you need to calculate values without the influence of particular dimensions, while still considering other dimensions in the view.





Standard



Show Me

Data

Analytics



Pages

Orders+ (sample_-supe...

Search



Tables

Orders

- Category
- City
- Country
- Customer ID
- Customer Name
- Order Date
- Order ID
- Order Profitable?
- Postal Code
- Product ID
- Product Name
- Region
- Row ID
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Avg sales per category
- Discount

Filters

Marks

Area



Columns

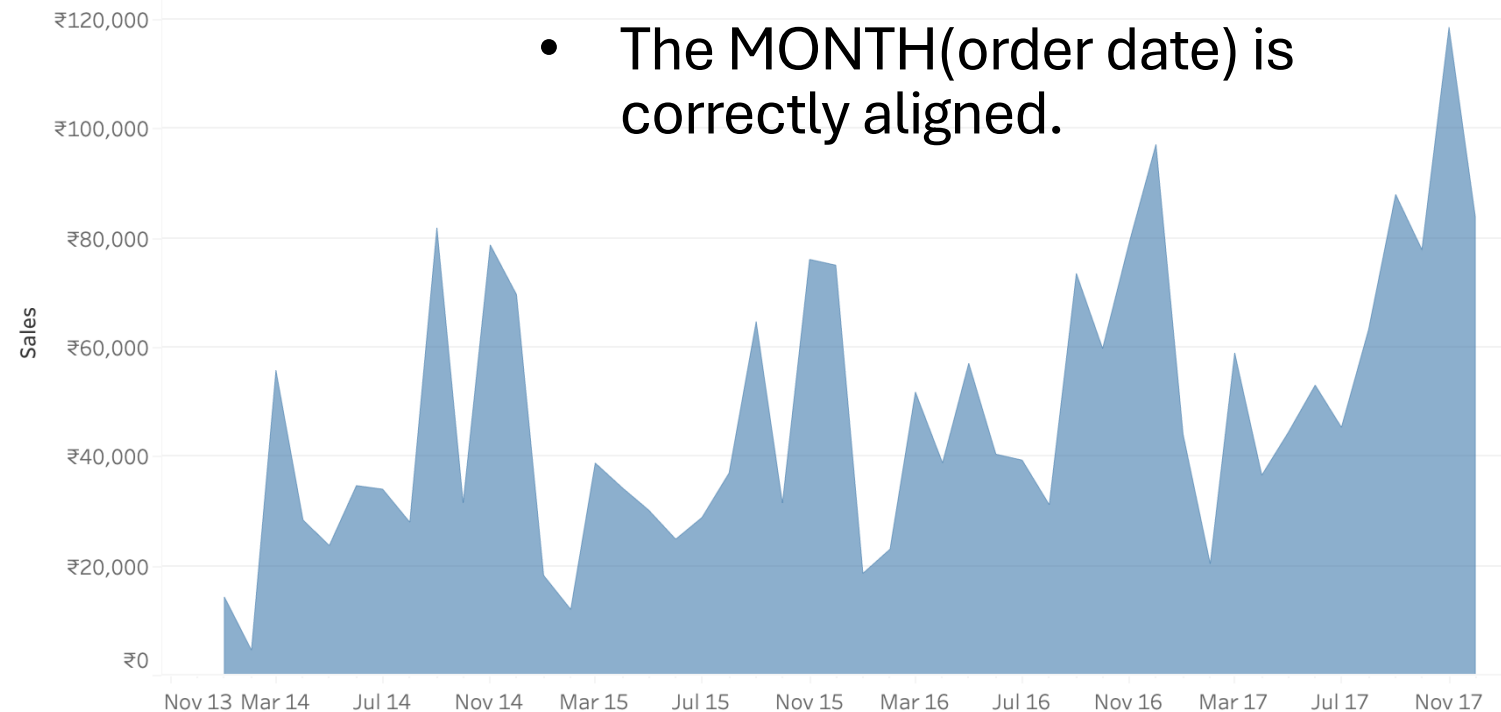
MONTH(Order Dat..

Rows

SUM(Sales)

Conversion of Dimension to Measure

Sales per Month (Order Date as Measure)



Data Source

Sheet 1

Sheet 2

Sheet 3

Sheet 4

Sheet 5

Sheet 6

Sheet 7

Sheet 8

Sheet 9

Sheet 10



48 marks 1 row by 1 column SUM(Sales): ₹2,297,201



Filters

Heat Map

Marks

☐ Automatic



Color



Size



Label



Detail



Tooltip



SUM(Profit)

SUM(Profit)



-5,300

12,790

Columns

Segment

Rows

Region

Sub-Category

Region	Sub-Catego..	Segment		
		Consumer	Corporate	Home Office
Central	Accessories			
	Appliances			
	Art			
	Binders			
	Bookcases			
	Chairs			
	Copiers			
	Envelopes			
	Fasteners			
	Furnishings			
	Labels			
	Machines			
	Paper			
	Phones			

Packed Bubble Chart

